SQL Exercise 2: Aggregate Function & Grouping

Query 1: Total of Employees (Count Function)

```
Select Count (*) As total_employees
From 'Workspace.default.employees';
```

Total Employees
10

Query 2: Find the total salary paid to all employees in the IT department. (Sum function)

```
Select Sum (Salary) As total_salary
From 'Workspace.default.employees_table
Where Department = 'IT';
```

Total Salary
220000

Query 3: Calculate the average salary of employees in the HR department. (AVG Function)

```
Select AVG (Salary) As average_salary
From 'Workspace.default.employees_table
Where department = 'HR';
```

Average_Salary
49500

Query 4: Find the highest and lowest salary in the company (Min() and Max() functions)

```
Select Min(Salary) As lowest_Salary,
       Max(Salary) As highest_Salary
From Workspace.default.employees;
```

Lowest Salary	Highest Salary
48000	62000

Query 5: To group employees by department & display the total salary paid in each department (Group By Statement)

```
Select department, Sum(Salary) As total_Salary
From 'Workspace.default.employees_table
Group By Department;
```

Department	Total_Salary
IT	220000
HR	99000
Finance	119000
Marketing	105000

Query 6: Count how many employees work in each city. (Group By & Count())

```
Select city, Count(*) As employees_Count
From 'Workspace.default.Employees_table
Group By City;
```

City	Employees_Count
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

Query 7: Group employees by department, calculate the average salary in each department, & order the results in descending order of average salary (Group By & Order By)

```
Select department, AVG(salary) As average_salary
From 'Workspace.default.employees_table'
Group By Department
Order By average_salary desc;
```

Department	AVG_Salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

Query 8: Find departments where the total salary paid exceeds 100,000 (Use GroupBy and Having)

```
Select department, Sum(Salary) As total_Salary
From 'Workspace.default.employees_table
Group By Department
Having Sum(Salary) > 100000
```

Department	Total_Salary
IT	220000
FINANCE	119000
MARKETING	105000

(Combining Group By, Having and Order By)

Query 9: List cities where more than one Employee works.

```
Select city, count(*) As employee_count
From 'Workspace.default.employees_table
Group By City
Having Count(*)>1
Order By employee_count Desc;
```

City	Employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

Query 10: Department with the highest Average Salary (Combining Aggregate Functions)

```
Select department, AVG(Salary) As average_salary
From 'Workspace.default.Employees_table
Group By department
Order By average_salary desc
Limit 1;
```

Department	Average_Salary
Finance	59500